

GhostHouses

Stage B Presentation

2025-2026-Spring Semester

02340312 - Yearly Project in Software Eng.-Stage B

GhostHouses Team

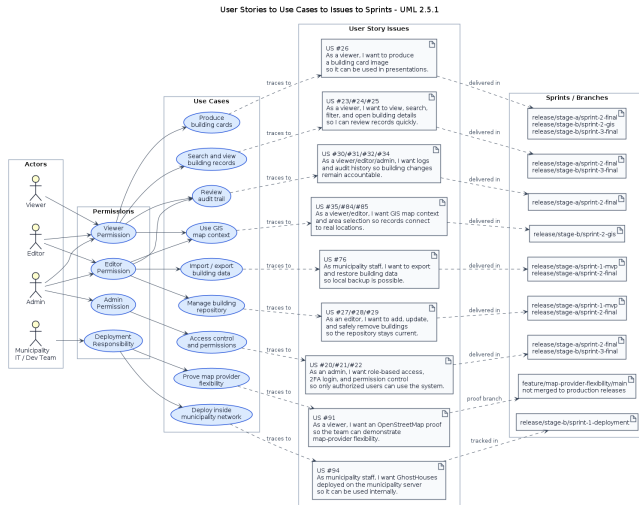
Technion, Faculty of Computer Science

Agenda

- 1 User Stories to Use Cases to Issues to Sprints
- 2 Architecture
- 3 Customer-Side Deployment

User Stories to Use Cases to Issues to Sprints

Traceability UML



User Stories to Use Cases to Issues to Sprints

- We grouped the backlog by actors, permission levels, and real user-facing use cases.
- Each use case maps to readable GitHub User Story issues, not only issue numbers.
- The diagram keeps implementation sub-issues out of the slide so the presentation stays readable.
- Release branches capture sprint checkpoints: Stage A MVP, Stage A final, Stage B deployment, Stage B GIS, and the final handoff state.

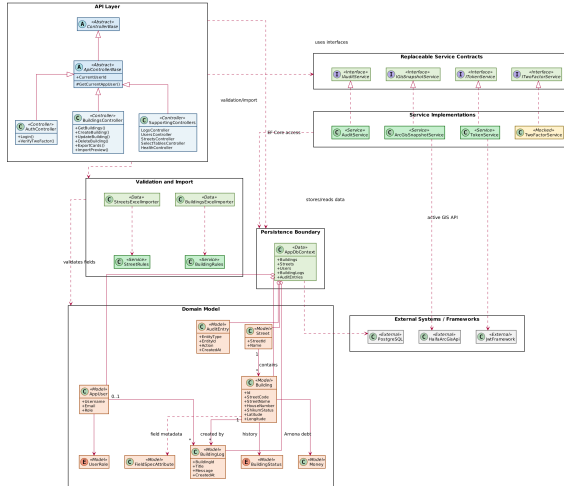
- **Branch management:** branches are organized around develop, feature branches, and release checkpoints.
- **Branch management:** feature work follows `develop → feature/<feature>/main → feature/<feature>/#<issue>-<slug> → feature/<feature>/main → develop`.
- **Issue management:** User Story issues describe user value and acceptance criteria.
- **Issue management:** implementation issues are linked to one parent User Story and one development branch.
- **Issue management:** closed implementation issues include a short completion comment with a Time spent line.

- **Automated tests:** GitHub Actions runs backend build/tests, frontend build, and Docker Compose build readiness.
- **Automated tests:** Issue Guard checks labels, milestone, project status, parent links, branches, and User Story acceptance criteria.
- **Readme.md:** the README is structured as a handoff guide with a clear Project Delivery section.
- **License:** the repository includes a Creative Commons Attribution 4.0 International license.

Architecture

UML Class Diagram

GhostHouses Maintainable Class Design - UML 2.5.1



Good Use of Abstract Classes and Design Patterns

- Abstract boundaries: service contracts for JWT, OTP, audit logging, and GIS snapshots.
- Concrete commitments: ASP.NET Core backend, React frontend, PostgreSQL database, ArcGIS GIS provider, and Docker Compose deployment.
- The external-risk areas stay behind interfaces, so OTP delivery or GIS provider changes do not require rewriting controllers or the domain model.
- Stable runtime choices are concrete because they define how the municipality receives and operates the system.

Good Use of Abstract Classes and Design Patterns

- `ApiControllerBase` centralizes authenticated API behavior.
- `BuildingRules` and `StreetRules` keep manual editing and Excel import aligned to one source of truth.
- `AppDbContext` is the persistence boundary for PostgreSQL entities.
- High cohesion: controllers, services, validation rules, import helpers, and models each have a clear responsibility.
- Low coupling: controllers depend on service contracts and stable boundaries instead of direct external systems.

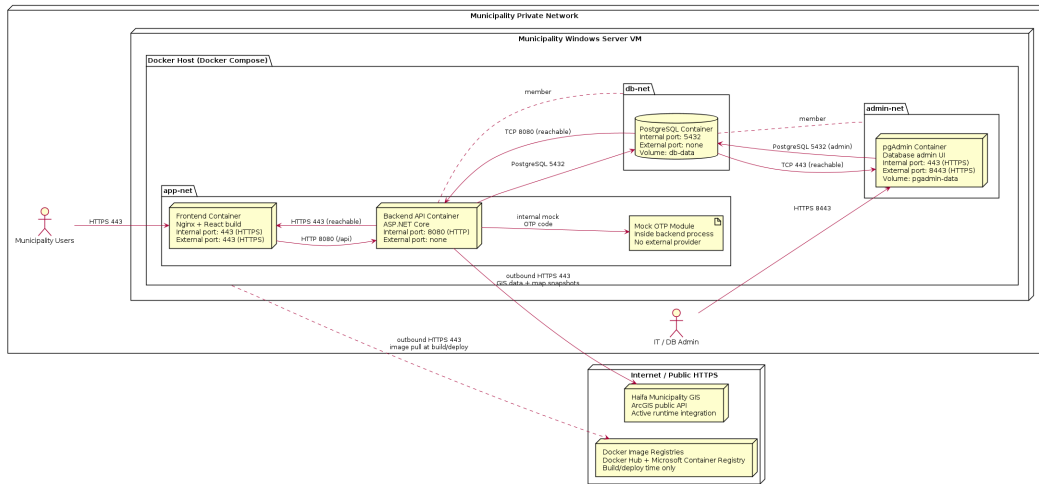
Robustness to Next Pivot / API Vendor Change

- **Future proof:** IGisSnapshotService protects building-card export from a future GIS vendor change.
- The delivered system uses Haifa Municipality ArcGIS because it is the client-approved provider.
- A separate OpenStreetMap proof branch demonstrates that the map provider can be replaced without changing the production delivery.
- OTP is mocked for handoff, but the flow depends on ITwoFactorService, so a real SMS/email provider can replace it later.
- Centralized business rules make future municipality field or validation changes safer.

Customer-Side Deployment

Deployment UML

GhostHouses Stage B Deployment Environment - UML 2.5.1



Customer-Side Deployment

- The municipality environment is a Windows Server VM inside the private municipality network.
- The intended deployment is a Docker Compose stack: frontend, backend, PostgreSQL, and pgAdmin.
- Municipality users access only the frontend through HTTPS 443.
- IT/database administration uses pgAdmin through HTTPS 8443.
- Backend and database stay internal to Docker networks and are not exposed directly.

Customer-Side Deployment Woes and Mitigation

- Main blocker: Docker Desktop / WSL 2 on the municipality Windows Server VM may require nested virtualization approval.
- We prepared the deployment as one repeatable command from `project/`: `docker compose down -v && docker compose up -d --build`.
- We documented ports, software dependencies, hardware expectations, secrets, certificates, and network access.
- We prepared fallback plans with the supervisor if nested virtualization is not approved.

Exhausting Communication With the Customer

- **Exhausting communication with the customer toward possible deployment, in writing:** deployment communication is documented with the client side, municipality dev team, IT/security contacts, and supervisors.
- The written trail covers deployment requirements, server preparation, security approval, and nested virtualization discussion.
- The client-side blocker is not application readiness, it is approval and infrastructure constraints in the municipality environment.
- We kept the system ready for handoff while continuing to coordinate professionally with the municipality side.

Thank You

GhostHouses Team